

A proposal of Mach-Zehnder Interferometer

Ong Kai Xuan, Nicholas (okaixuan@dso.org.sg)

DSO National Laboratories, Singapore

Abstract: This report describes the proposed design of an imbalanced Mach-Zehnder interferometer (MZI) to extract the waveguide group index. The aim of the proposal is to investigate the sensitivity of the interferometer to manufacturing variabilities. To do this, multiple interferometers of the same design will be fabricated and their extracted waveguide group indexes (n_g) will be analysed. The results are expected to show the effectiveness of the school's waveguide manufacturing process and provide insights on the impact of manufacturing variabilities on photonics integrated circuits.

1. Introduction

On November 2022, OpenAI released ChatGPT to the public [1] and since then, the technology known as Artificial Intelligence (AI) has skyrocketed in popularity. In the years to come, many companies participated in this AI race in hopes to take the throne of AI technology. One of the methods includes the investment and further research into integrated photonics due to their high bandwidth capabilities, energy efficiency and scalability [2]. The Mach-Zehnder Interferometer (MZI) is one of these essential photonics components for building integrated photonics as it is able to help extract waveguide group indexes (n_g) which is an important waveguide parameter that influences the spectral response of photonics devices.

However, the capability of a MZI to accurately extract waveguide group indexes can be affected by manufacturing variabilities. Even small discrepancies in waveguide dimensions can lead to a slight difference from desired n_g which will then affect the waveguide's free spectral range (FSR), influencing the whole operation range of the integrated photonics circuit.

Therefore, the aim of this proposal is to design a standard imbalanced MZI and go through several fabrication of the same design to observe and investigate the importance of manufacturing variability.

The organisation of this report will be as follows:

Chapter 1, an introduction into project aim and motivations

Chapter 2, the theory behind how a MZI works

Chapter 3, modelling and simulation of the proposed imbalanced MZI design. Along with its expected results.

Chapter 4, a conclusion on the proposal and discussion of future works.

2. Theory – MZI

An imbalanced MZI starts with a beam splitter. For example, a Y-branch splitter with an input intensity, I_i , and electric field, E_i . The light will first be split equally between the two branches and then combined at the end with a Y-branch combiner. At the first split, because intensity is equal, I_1 is equal to I_2 which is equal to $I_i/2$. Same goes for the electric field except E_1 is equal to E_2 which is equal to $E_i/\sqrt{2}$ as I_i is directly proportional to $|E_i|^2$. [3]

At the second split, the Y-branch combiner as an example, the output field will be a vector summation of the input fields which are E_1 and E_2 . If a plane wave is used, the electric field will be $E = E_0 e^{i(\omega t - \beta z)}$ where the propagation constant $B = (2\pi n/\lambda)$ and n is the index of refraction. [3] For a lossless case and the length of the branches being imbalanced, L_1 and L_2 , we have

$$E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} \quad (1)$$

$$E_{o2} = E_2 e^{-i\beta L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} \quad (2)$$

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta L_1} + e^{-i\beta L_2}) \quad (3)$$

The intensity I_o will be therefore

$$I_o = \frac{I_i}{4} |e^{-i\beta L_1} + e^{-i\beta L_2}|^2 = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (4)$$

Additionally, considering that it is an imbalanced MZI, FSR can be found using

$$FSR = \frac{\lambda^2}{\Delta L n_g} \quad (5)$$

Where n_g refers to the group index

$$n_g = n - \lambda \frac{dn}{d\lambda} \quad (6)$$

3. Modelling and Simulation

The imbalanced MZI will be modelled using ANSYS Lumerical INTERCONNECT software. Following the example in chapter, the MZI will be made using two Y-branches, a splitter and combiner, with strip waveguides connected for each of the split branches. The strip waveguides will have a design parameter width of 500nm width, a height of 220nm and a

length difference of $100\mu\text{m}$ between the two branches. The materials used for the waveguide will be a silicon channel in a silicon dioxide substrate.

The waveguide simulations, Figure 1 and Figure 2, show that the waveguide is mainly transverse electric field (TE) polarised.

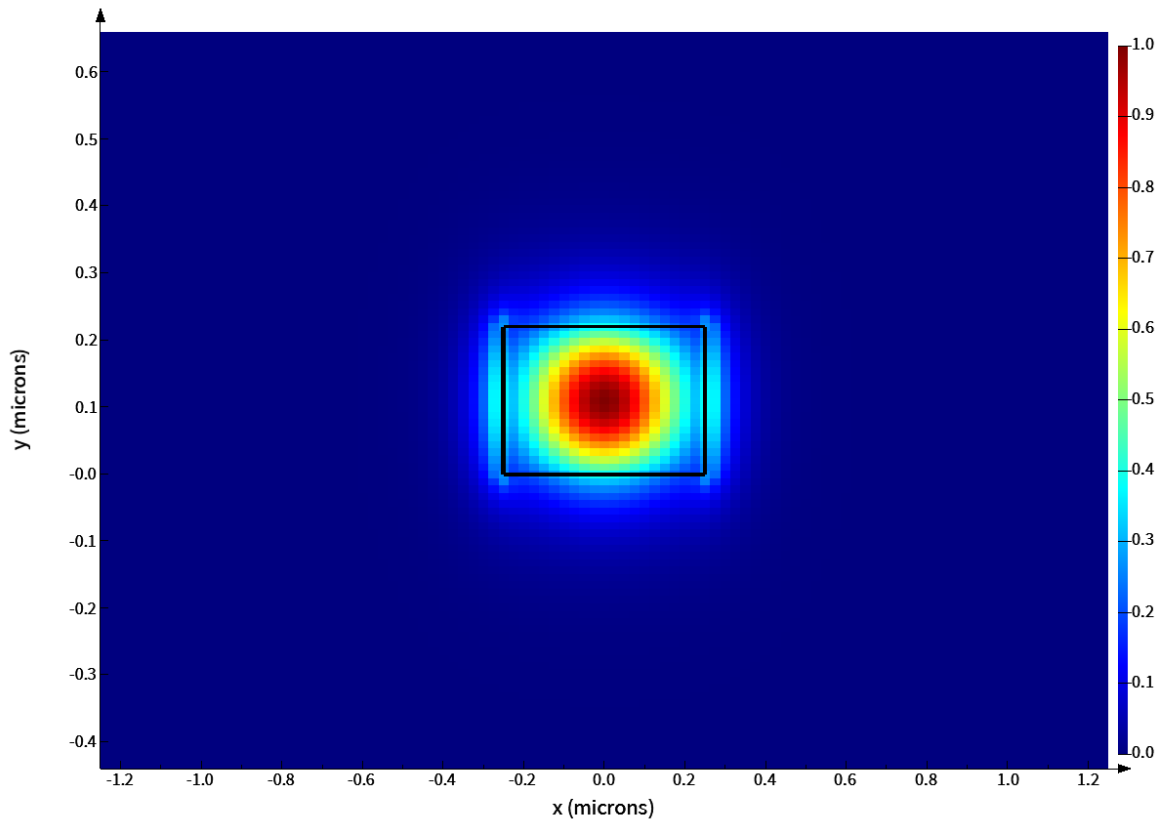


Figure 1: TE field of waveguide

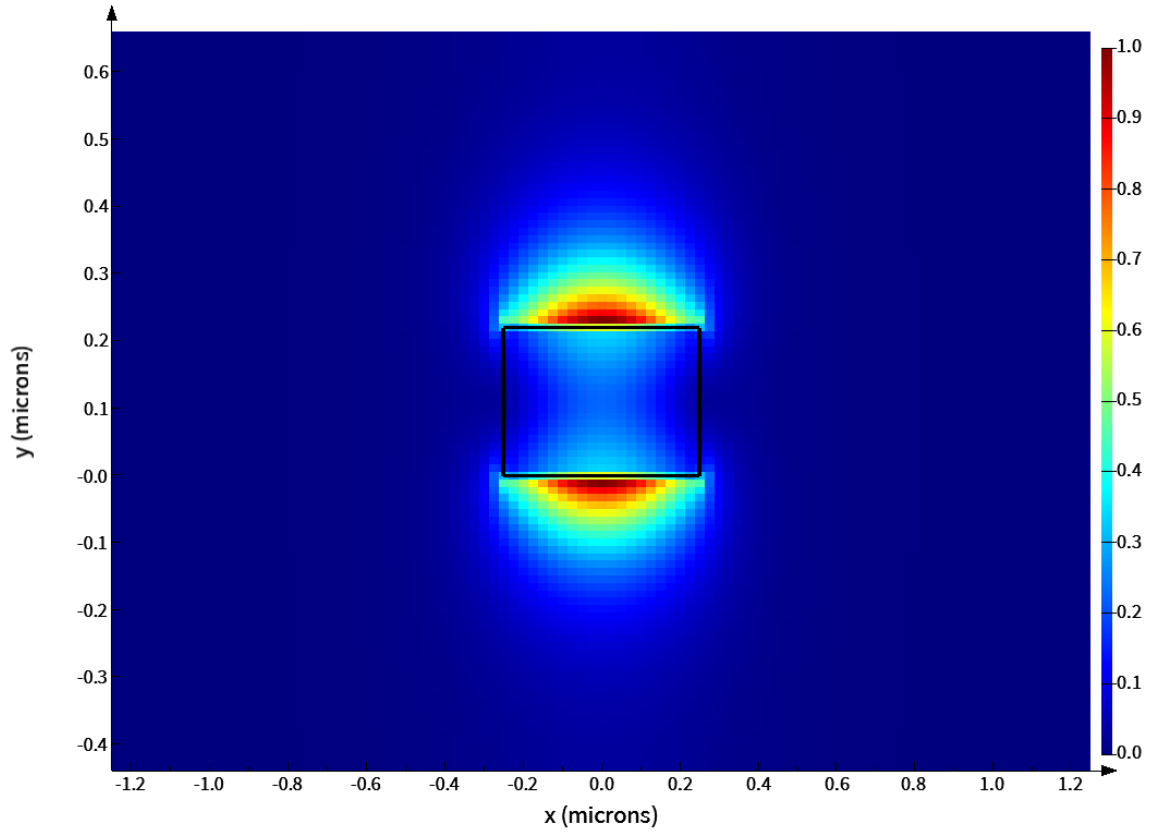


Figure 2: TM field of waveguide

Figure 3 and figure 4 shows the effective index (n_{eff}) and the group index, n_g , of the waveguide as a function of wavelength. This shows that n_{eff} decreases with increasing wavelength and n_g increases with wavelength.

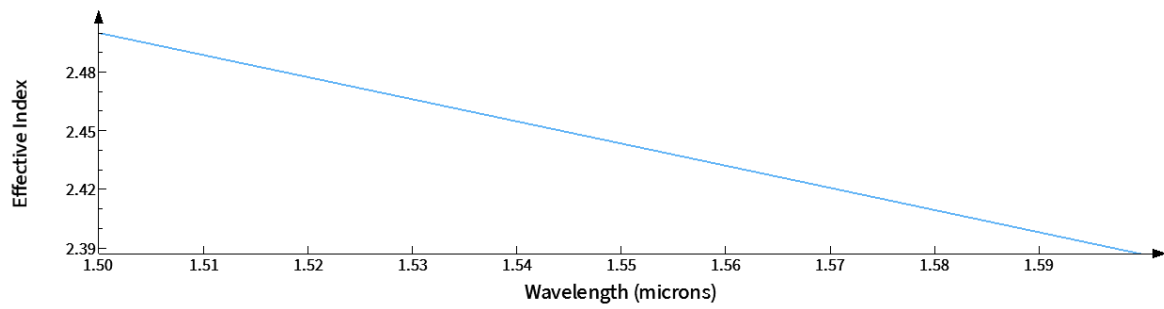


Figure 3: n_{eff} against wavelength(μm)

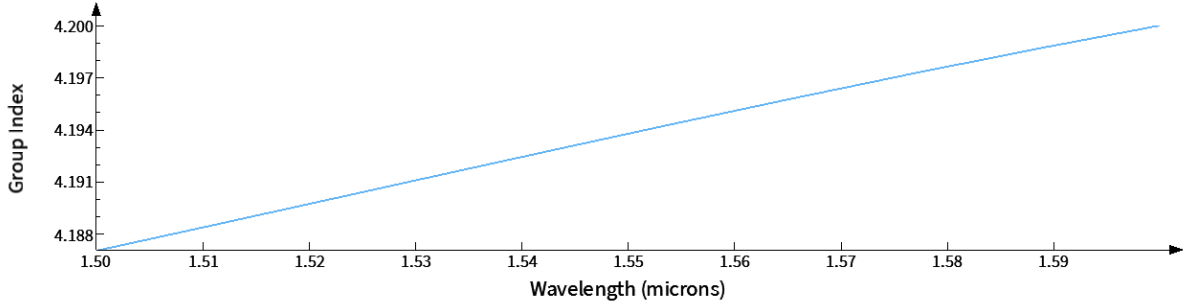


Figure 4: n_g against wavelength(μm)

After obtaining the wavelength sweep, the data can be used to calculate the compact model of the waveguide

$$n_{eff}(\lambda) = 2.44 - 1.13(\lambda - 1.55) - 0.04(\lambda - 1.55)^2 \quad (7)$$

Regarding the FSR of the waveguide, we can also observe that as ΔL and wavelength used is constant, $\Delta L=100\mu\text{m}$ and $\lambda=1550\text{nm}$, FSR can be used to extract the value of the group index, n_g . This is important as through the FSR value, we can observe the seriousness of the manufacturing variability. If the FSR and the extracted value of n_g is not the desired one, it will lead to

$$\text{Optical delay difference} = \frac{n_g \Delta L}{c} \quad (8)$$

Manufacturing variability	FSR (nm)	n_g
None	5.6747	4.235
$\Delta L=105\mu\text{m}$	5.4727	4.182
Waveguide width= $0.4\mu\text{m}$	5.3972	4.451

Table 1: Effects of Manufacturing variability on FSR and then on n_g

4. Conclusion

In this paper, we propose to design a simple imbalanced MZI with Y-branch splitter and combiner. With its fixed parameters, it is to be fabricated 10 times and measured accordingly in order to observe the severity of manufacturing variability. The simulations given in this paper will be used as a measure of control for the observation.

References

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- [2] L. Torrijos-Morán and D. Pérez-López, “Industry insight: photonics to scale AI data centers,” *Npj Nanophotonics*, vol. 3, no. 1, p. 8, Jan. 2026, doi: 10.1038/s44310-025-00105-1.
- [3] L. Chrostowski and M. Hochberg, “Silicon Photonics Design: From Devices to Systems”, Cambridge University Press, 2015.